

REACTJS

- **React Introduction**

- Overview of frameworks, libraries for client side Web applications
- React introduction
- Environment Setup for React Application
- Understanding NPM commands
- VS Code extensions for ES6, React

- **React Essential Features and Syntax**

- React App Project Directory Structure
- Overview of Webpack, Babel
- React Component Basic
- Create React Component
- Understanding JSX
- Limitations of JSX
- Working with Components and Reusing Components

- **React Components, Props and State**

- Understanding and using Props and State
- Handling Events with methods
- Manipulating the State
- Two way data-binding
- Functional (Stateless) VS Class (Stateful) Components
- Parent – Child Communication
- Dynamically rendering contents
- Showing Lists, List and keys

- **Styling Components**

- CSS Styling
- Scoping Styles using Inline Styles
- Limitations of inline styles
- Inline Styles with Radium
- Using Psuedo classes/media queries with inline styles
- CSS Modules, importing css classes
- Adding Bootstrap, Semantic UI to React apps
- Using react-bootstrap, reactstrap packages

- **Debugging React Apps**

- Understanding React Error Messages
- Handling Logical Errors,
- Debugging React apps using google developer tools and React DevTool
- Understanding Error Boundaries

- **React Component life cycle**

- Updating life cycle hooks
- PureComponent
- React's DOM Updating Strategy
- Returning adjacent elements
- Fragments

- **React Component in Details**

- Higher Order Components
- Passing unknown Props
- Validating Props
- Using References
- React Context API
- Updated Lifecycle hooks (16.3)
- Best practices for React Projects
- Demo apps

- **HTTP Requests/Ajax Calls**

- HTTP Requests in React
- Introduction of Axios package
- HTTP GET Request, fetching & transforming data

- HTTP POST, DELETE, UPDATE

- Handling Errors
- Adding/Removing Interceptors
- Creating/Using Axios instances

- **React Routing**

- Routing and SPAs
- Setting Up the Router Package
- react-router vs react-router-dom
- Preparing the Project For Routing
- Switching Between Pages, Routing-Related Props
- The "withRouter" HOC & Route Props
- Passing & extracting route/query parameters

- Using Switch to Load a Single Route
- Navigating Programmatically

- **React Forms and Form Validation**

- Creating a Custom Dynamic Input Component
- Setting Up a JS Config for the Form
- Dynamically Create Inputs based on JS Config
- Adding a Dropdown Component
- Handling User Input
- Handling Form Submission
- Adding Custom Form Validation
- Fixing a Common Validation
- Adding Validation Feedback
- Showing Error Messages
- Handling Overall Form Validity
- Deploying React App to the Web
- Testing React apps with JEST